

Nick Nunez

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Summary

Self-taught software engineer with a professional background in production web applications and a growing focus on systems programming in C/C++. Personal projects explore what it means to write performance-critical, real-time software close to the platform — deliberately avoiding high-level audio frameworks in favor of working directly with macOS system APIs to build a deeper understanding of how these systems actually work. Bring strong instincts for API design, library architecture, and clean separation of concerns developed across both professional and personal work.

Projects

Standalone Groovebox for macOS

github.com/nico-nunez

- Standalone C/C++ performance instrument combining a shared clock, pattern-based step sequencer, and multi-track synthesis — built from scratch without audio framework abstractions
- Integrated directly with CoreAudio and CoreMIDI (no JUCE or RtAudio) to understand the macOS audio platform at the system API level: audio device setup, buffer callbacks, MIDI event collection, and thread scheduling
- Embedded a Lua runtime as a live REPL control surface, enabling real-time parameter changes, sequencer control, and performance scripting without recompiling
- Step sequencer supports per-step parameter locks; multi-track architecture with internal mixing and shared clock synchronization across parts
- Hosts the meh-synth-engine library; built-in code editor with language server support

C/C++, CoreAudio, CoreMIDI, Lua, Make, macOS

Embeddable C/C++ Synthesizer Library

github.com/nico-nunez

- Standalone C/C++ library extracted from the groovebox application — builds as a static lib and can be embedded into other projects via `engine.mk`; designed with a clean host/engine boundary so the engine owns synthesis state and the host owns platform concerns
- Implements 4 wavetable oscillators per voice, 64-voice polyphony, 3 envelope generators, 3 LFOs with tempo-sync and fade-in, and a 16-slot modulation matrix with sources including envelopes, LFOs, velocity, mod wheel, and key tracking

- Renders in internal 64-sample chunks regardless of host buffer size, providing a cheap control-rate lane; parameters use a dirty-sync model to defer coefficient recalculation rather than recomputing on every write
- Preset system captures full engine state — scalar params, FM routing, modulation matrix, signal and FX chain ordering — with JSON serialization for persistence
- Event-driven API surfaces three input channels: MIDI events, scalar parameter updates, and higher-level engine commands

C++17, Make, static library, JSON

Experience

Software Engineer | Publicis Media

September 2022 – November 2025

- Progressed from individual contributor to technical lead in under 3 years; owned hiring of junior and mid-level developers, conducted code reviews, and established engineering best practices across the team
- Built and maintained production TypeScript/React applications processing 10–50K records per session — designed component architecture, optimized performance through parallel requests and client-side caching, and wrote test coverage with Vitest and React Testing Library
- Contributed to Django/Python backend services — designed API contracts, identified and diagnosed issues across a large Django codebase, and collaborated directly with the data engineering team on pipeline integration
- Reverse-engineered undocumented Java/Spring Boot APIs from source to unblock frontend integration; identified and led the replacement of insecure API access patterns with JWT authentication

Lawn Care Technician | TruGreen

2013 – 2021

- Advanced from residential to commercial accounts over 8 years; transitioned into software engineering through self-directed study

Technical Skills

Systems: C/C++ (C++17), Make, static libraries, real-time programming

Platform APIs: CoreAudio, CoreMIDI, macOS system frameworks

Scripting / Embedding: Lua (embedded runtime), shell scripting

Web / Professional: TypeScript, React, Python, Django, Node.js

Database: PostgreSQL, Drizzle ORM, SQL

Testing: Vitest, Playwright, React Testing Library

Cloud & Infra: AWS (EC2, S3), GCP, Docker

Tools: Git, Vite, Docker Compose, Jenkins

Education

High School Diploma